

Assignment (Unit-3)

SECTION A (5 questions * 2 marks = 10 Marks)

1. Define IoT security and mention two common vulnerabilities in IoT devices.
2. What are the risks of using public Wi-Fi networks?
3. Define remote working security and list two major threats associated with it.
4. What is data disposal? Why is it important in cybersecurity?
5. What is multi-factor authentication (MFA)? How does it improve security?

SECTION B (3 questions * 7 marks = 21 Marks)

1. A user installs multiple IoT devices at home. Analyze the security risks, explain how attackers exploit such devices, and suggest countermeasures.
2. An employee accesses company data using public Wi-Fi. Explain the risks, attack methods used by hackers, and best practices to secure remote access.
3. Explain the importance of secure data disposal by discussing risks of improper disposal and techniques used to securely erase data.

SECTION C (3 questions * 11 marks = 33 Marks)

1. A smart home environment includes devices like Alexa, smart TV, and IoT appliances. Design a complete IoT security framework including threat mitigation and privacy protection strategies.
2. Discuss remote working security in detail by explaining risks, secure communication practices, and guidelines for safe remote access to organizational resources.
3. Before selling old computers, an organization must ensure data safety. Explain secure data deletion techniques and propose a standard operating procedure for safe device disposal.